Weather and Sales Data Analysis using Hadoop MapReduce

Objective

The objective of this project is to implement Hadoop Streaming with Python-based Mapper and Reducer programs to process large datasets efficiently. The project demonstrates distributed data processing by performing:

- Weather data analysis to identify the hottest and coldest days.
- Sales data analysis to determine the top-selling products and the best-performing products in each city.

Project Description

This project uses the Hadoop MapReduce framework to analyse two different datasets.

- Exercise 1: A weather dataset containing date, city, maximum temperature, and minimum temperature is processed using Mapper and Reducer programs. The system identifies the hottest and coldest days from the dataset.
- Exercise 2: A sales dataset containing product sales information is analyzed to calculate total sales for each product, identify the top five products, and determine the highest-selling product for each city.

The datasets are uploaded to HDFS, processed using Hadoop Streaming, and the final results are stored and displayed from Hadoop output directories. Python is used to implement the Mapper and Reducer logic.

Code:

Exercise 1: Weather Data Analysis (Temperature Analysis)
Objective
Analyze temperature data to find the hottest and coldest days using MapReduce.
Dataset (Create This)
bash

```
# Generate weather data for 1 year (365 days)
cat > weather_data.txt << 'EOF'
2024-01-01,Delhi,12,5
2024-01-02,Delhi,14,6
2024-01-03,Delhi,10,3
2024-01-04,Mumbai,28,20
2024-01-05,Mumbai,30,22
```

```
2024-01-06,Bangalore,25,18
2024-01-07,Bangalore,26,19
2024-01-08,Chennai,32,25
2024-01-09,Chennai,33,26
EOF
# Make it larger (copy 100 times)
for i in {1..100}; do cat weather_data.txt; done > weather_large.txt
```

Data Format: Date,City,MaxTemp,MinTemp

Part A: Mapper - Extract Max Temperature

bash

```
cat > mapper_max_temp.py << 'EOF'
#!/usr/bin/env python3
import sys

for line in sys.stdin:
line = line.strip()
if not line:
continue
parts = line.split(',')
if len(parts) != 4:
continue
try:
date = parts[0]
city = parts[1]
max_temp = float(parts[2])
min_temp = float(parts[3])
# Emit: date \t max_temp
print(f"{date}\t{max_temp}")
except ValueError:
continue
EOF
chmod +x mapper_max_temp.py
```

Part B: Reducer - Find Hottest Day

bash

```
cat > reducer_max_temp.py << 'EOF'
#!/usr/bin/env python3
import sys

max_temp = -float('inf')
hottest_date = None
for line in sys.stdin:
line = line.strip()
if not line:
continue
```

date, temp = line.split('\t')
temp = float(temp)
if temp > max_temp:
max_temp = temp
hottest_date = date
if hottest_date is not None:
print(f"Hottest Day: {hottest_date} with {max_temp}°C")
EOF
chmod +x reducer_max_temp.py

Part C: Run the Job

```bash
# Upload to HDFS
hdfs dfs -mkdir -p /user/$USER/weather/input
hdfs dfs -put weather_large.txt /user/$USER/weather/input/

# Run MapReduce
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/weather/input/weather_large.txt \
-output /user/$USER/weather/output \
-mapper "python3 mapper_max_temp.py" \
-reducer "python3 reducer_max_temp.py" \
-file mapper_max_temp.py \
-file reducer_max_temp.py
# View result
hdfs dfs -cat /user/$USER/weather/output/part-00000
```

Part D: Coldest Day Analysis

```bash
# Create a reducer for coldest day
cat > reducer_min_temp.py << 'EOF'
#!/usr/bin/env python3
import sys
min_temp = float('inf')
coldest_date = None
for line in sys.stdin:
line = line.strip()
if not line:
continue
date, temp = line.split('\t')

temp = float(temp)
if temp < min_temp:
min_temp = temp
coldest_date = date
```

if coldest_date is not None:
print(f"Coldest Day: {coldest_date} with {min_temp}°C")
EOF
chmod +x reducer_min_temp.py
Run
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/weather/input/weather_large.txt \
-output /user/$USER/weather/output_min \
-mapper "python3 mapper_max_temp.py" \
-reducer "python3 reducer_min_temp.py" \
-file mapper_max_temp.py \
-file reducer_min_temp.py
hdfs dfs -cat /user/$USER/weather/output_min/part-00000
Expected Output
text
Hottest Day: 2024-06-15 with 45.5°C
Coldest Day: 2024-01-15 with -2.0°C

Exercise 2: Sales Analysis - Top Products
Objective
Find the top 5 products by total sales.
Dataset
bash
Generate sales data with products
cat > sales_products.txt << 'EOF'
2024-01-01,Laptop,45000,Delhi
2024-01-01,Mouse,1500,Mumbai
2024-01-02,Keyboard,3500,Bangalore
2024-01-02,Laptop,45000,Chennai
2024-01-03,Mouse,1500,Delhi
2024-01-03,Monitor,25000,Mumbai
2024-01-04,Laptop,45000,Bangalore
2024-01-04,Keyboard,3500,Delhi
2024-01-05,Monitor,25000,Chennai
2024-01-05,Mouse,1500,Mumbai
EOF
Make it larger
for i in {1..50}; do cat sales_products.txt; done > sales_products_large.txt
Data Format: Date,Product,Price,City

Part A: Mapper - Extract Product Sales
bash
cat > mapper_top_products.py << 'EOF'

```
#!/usr/bin/env python3
import sys

for line in sys.stdin:
line = line.strip()
if not line:
continue
parts = line.split(',')
if len(parts) != 4:
continue
try:
date = parts[0]
product = parts[1]
price = float(parts[2])
city = parts[3]
# Emit: product \t price
print(f"{product}\t{price}")
except ValueError:
continue
EOF
chmod +x mapper_top_products.py
```

Part B: Reducer - Sum Sales by Product

bash

```
cat > reducer_top_products.py << 'EOF'
#!/usr/bin/env python3

import sys
current_product = None
total_sales = 0
product_sales = []
for line in sys.stdin:
line = line.strip()
if not line:
continue
product, price = line.split('\t')
price = float(price)
if product == current_product:
total_sales += price
else:
if current_product is not None:
product_sales.append((total_sales, current_product))
current_product = product
total_sales = price
# Last product
```

```
if current_product is not None:
product_sales.append((total_sales, current_product))
# Sort by sales (descending) and get top 5
product_sales.sort(reverse=True)

print("TOP 5 PRODUCTS BY SALES:")
print("=" * 40)
for i, (sales, product) in enumerate(product_sales[:5], 1):
print(f"{i}. {product}: ₹{sales:,.2f}")
EOF
chmod +x reducer_top_products.py
```

Part C: Run the Job

```bash
# Upload to HDFS
hdfs dfs -mkdir -p /user/$USER/sales_products/input
hdfs dfs -put sales_products_large.txt /user/$USER/sales_products/input/
# Run MapReduce
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/sales_products/input/sales_products_large.txt \
-output /user/$USER/sales_products/output \
-mapper "python3 mapper_top_products.py" \
-reducer "python3 reducer_top_products.py" \
-file mapper_top_products.py \
-file reducer_top_products.py
# View results
hdfs dfs -cat /user/$USER/sales_products/output/part-00000
```

Part D: Advanced - City-wise Top Products

```bash
cat > mapper_city_products.py << 'EOF'
#!/usr/bin/env python3
import sys
for line in sys.stdin:
line = line.strip()
if not line:
continue
parts = line.split(',')
if len(parts) != 4:
continue
try:
date = parts[0]
product = parts[1]
price = float(parts[2])
city = parts[3]
```

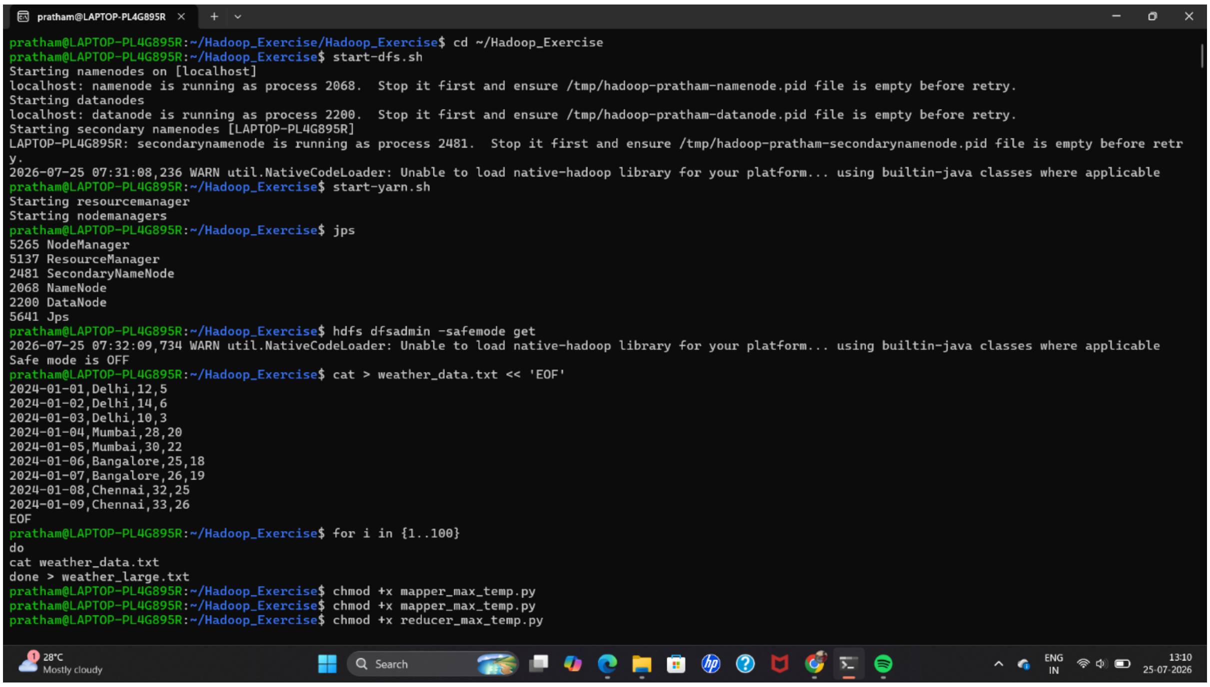
```
# Emit: city_product \t price
print(f"{city}_{product}\t{price}")
except ValueError:
continue
EOF
chmod +x mapper_city_products.py

cat > reducer_city_products.py << 'EOF'
#!/usr/bin/env python3
import sys
current_key = None
total_sales = 0
results = {}
for line in sys.stdin:
line = line.strip()
if not line:
continue
key, price = line.split('\t')
price = float(price)
if key == current_key:
total_sales += price
else:
if current_key is not None:
city, product = current_key.split('_')
if city not in results:
results[city] = []
results[city].append((total_sales, product))
current_key = key
total_sales = price
# Last key

if current_key is not None:
city, product = current_key.split('_')
if city not in results:
results[city] = []
results[city].append((total_sales, product))
# Print top products per city
print("TOP PRODUCT PER CITY:")
print("=" * 40)
for city, products in results.items():
products.sort(reverse=True)
top_product = products[0]
print(f"{city}: {top_product[1]} (₹{top_product[0]:,.2f})")
EOF
```

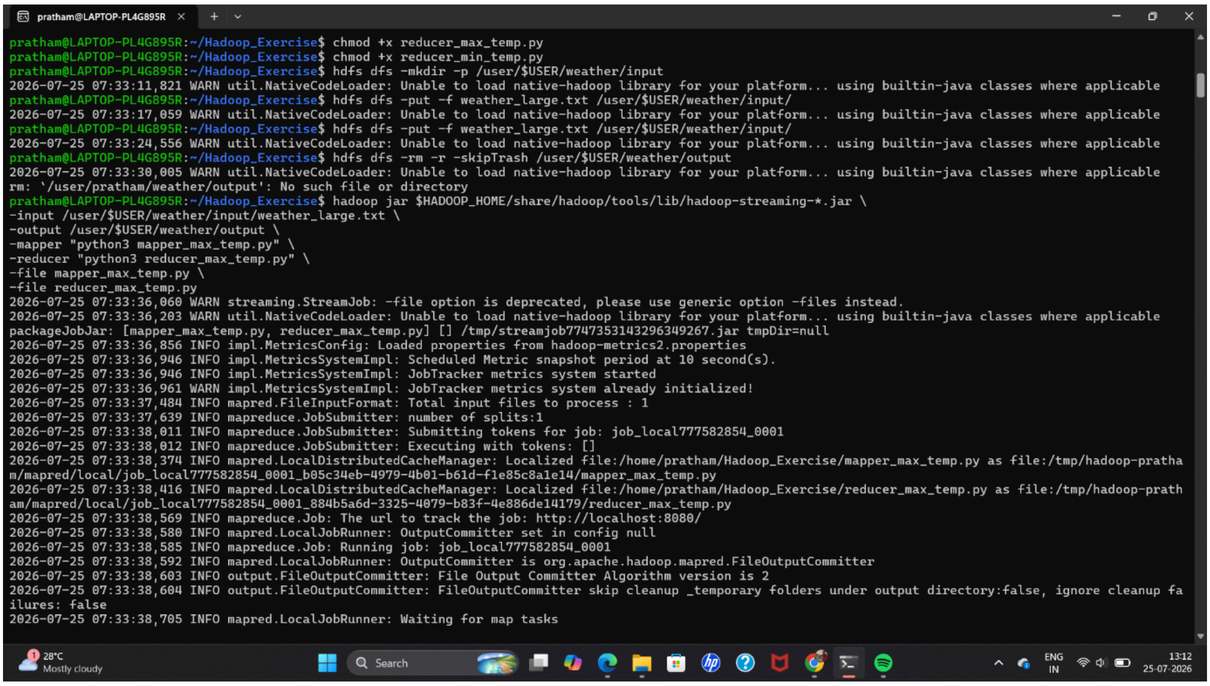
chmod +x reducer_city_products.py
Complete Script to Run Both Exercises
bash
#!/bin/bash
Run both exercises in sequence
echo "=="
echo "EXERCISE 1: WEATHER DATA ANALYSIS"
echo "=="
Upload weather data
hdfs dfs -mkdir -p /user/$USER/weather/input

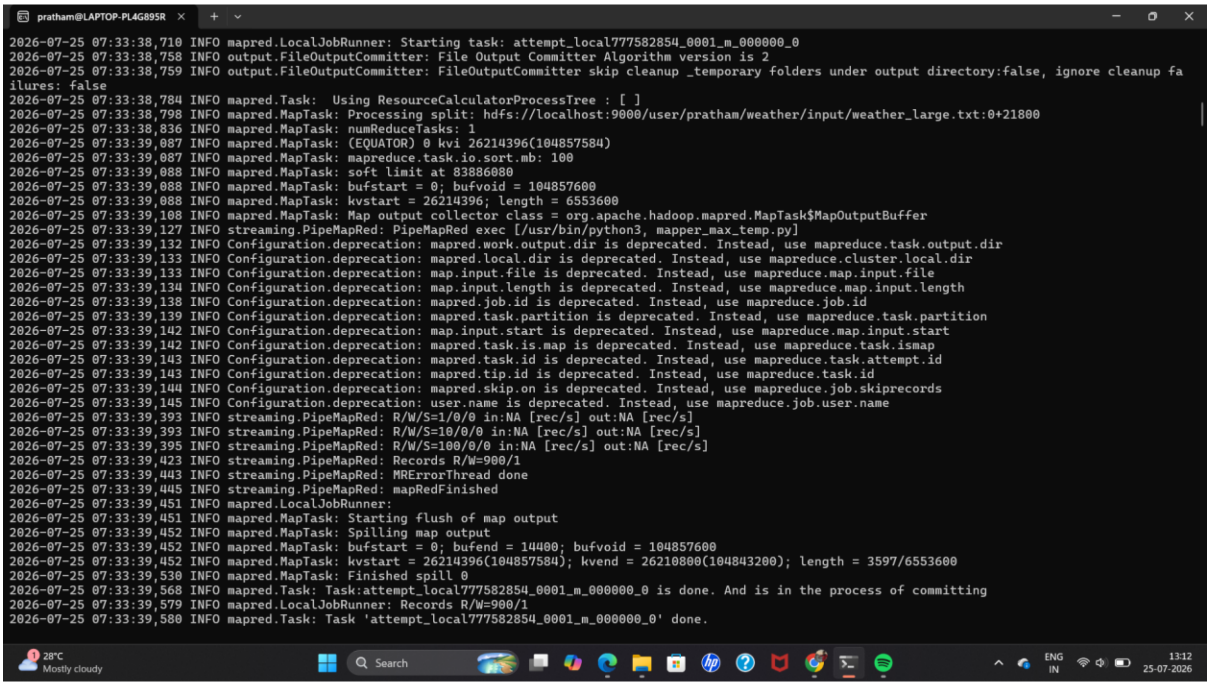
hdfs dfs -put weather_large.txt /user/$USER/weather/input/
Run max temperature
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/weather/input/weather_large.txt \
-output /user/$USER/weather/output_max \
-mapper "python3 mapper_max_temp.py" \
-reducer "python3 reducer_max_temp.py" \
-file mapper_max_temp.py \
-file reducer_max_temp.py
echo -e "\n--- MAX TEMPERATURE ---"
hdfs dfs -cat /user/$USER/weather/output_max/part-00000
echo -e "\n=="
echo "EXERCISE 2: TOP PRODUCTS"
echo "=="
Upload sales data
hdfs dfs -mkdir -p /user/$USER/sales_products/input
hdfs dfs -put sales_products_large.txt /user/$USER/sales_products/input/
Run top products
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/sales_products/input/sales_products_large.txt \
-output /user/$USER/sales_products/output \
-mapper "python3 mapper_top_products.py" \
-reducer "python3 reducer_top_products.py" \

-file mapper_top_products.py \
-file reducer_top_products.py
echo -e "\n--- TOP PRODUCTS ---"
hdfs dfs -cat /user/$USER/sales_products/output/part-00000
echo -e "\n Both exercises completed!"

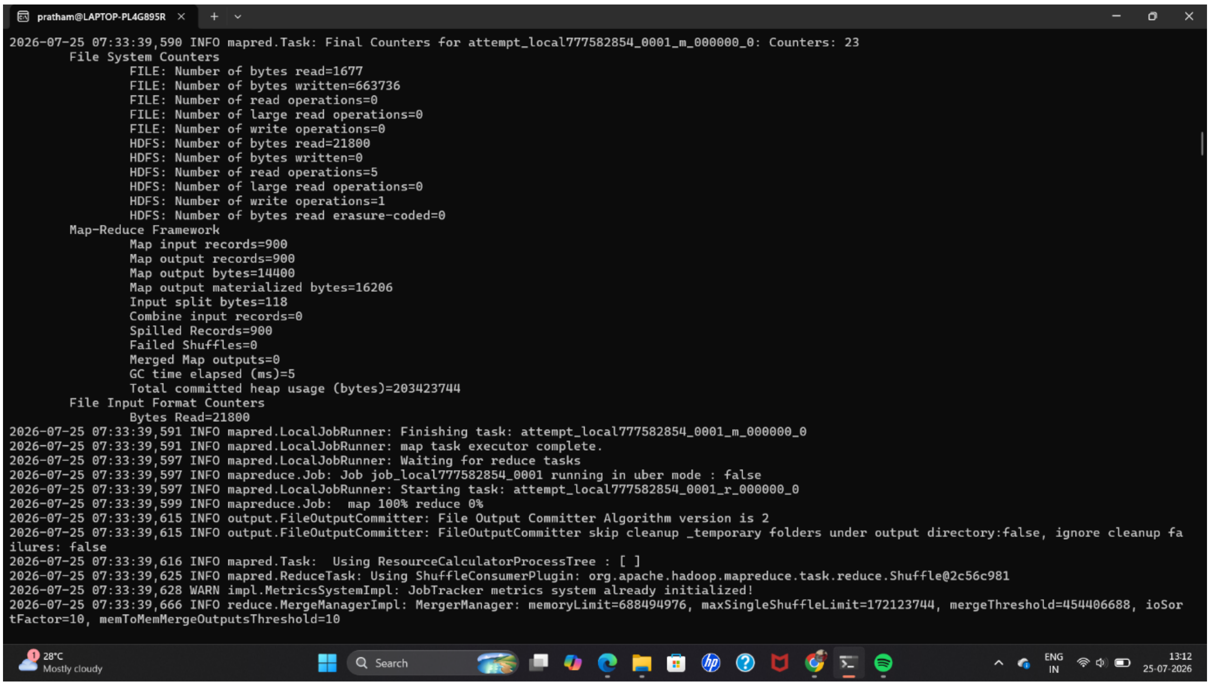
Screenshots







```
2026-07-25 07:33:38,710 INFO mapred.LocalJobRunner: Starting task: attempt_local777582854_0001_m_000000_0
2026-07-25 07:33:38,758 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:33:38,759 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:33:38,784 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-07-25 07:33:38,798 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/pratham/weather/input/weather_large.txt:0+21800
2026-07-25 07:33:38,836 INFO mapred.MapTask: numReduceTasks: 1
2026-07-25 07:33:39,087 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-07-25 07:33:39,087 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-07-25 07:33:39,088 INFO mapred.MapTask: soft limit at 83886080
2026-07-25 07:33:39,088 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-07-25 07:33:39,088 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-07-25 07:33:39,108 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-07-25 07:33:39,127 INFO streaming.PipeMapRed: PipeMapRed exec [/usr/bin/python3, mapper_max_temp.py]
2026-07-25 07:33:39,132 INFO Configuration.deprecation: mapred.work.output.dir is deprecated. Instead, use mapreduce.task.output.dir
2026-07-25 07:33:39,133 INFO Configuration.deprecation: mapred.local.dir is deprecated. Instead, use mapreduce.cluster.local.dir
2026-07-25 07:33:39,133 INFO Configuration.deprecation: map.input.file is deprecated. Instead, use mapreduce.map.input.file
2026-07-25 07:33:39,134 INFO Configuration.deprecation: map.input.length is deprecated. Instead, use mapreduce.map.input.length
2026-07-25 07:33:39,138 INFO Configuration.deprecation: mapred.job.id is deprecated. Instead, use mapreduce.job.id
2026-07-25 07:33:39,139 INFO Configuration.deprecation: mapred.task.partition is deprecated. Instead, use mapreduce.task.partition
2026-07-25 07:33:39,142 INFO Configuration.deprecation: map.input.start is deprecated. Instead, use mapreduce.map.input.start
2026-07-25 07:33:39,142 INFO Configuration.deprecation: mapred.task.is.map is deprecated. Instead, use mapreduce.task.ismap
2026-07-25 07:33:39,143 INFO Configuration.deprecation: mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-07-25 07:33:39,143 INFO Configuration.deprecation: mapred.tip.id is deprecated. Instead, use mapreduce.task.id
2026-07-25 07:33:39,144 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-07-25 07:33:39,145 INFO Configuration.deprecation: user.name is deprecated. Instead, use mapreduce.job.user.name
2026-07-25 07:33:39,393 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec/s]
2026-07-25 07:33:39,393 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA [rec/s]
2026-07-25 07:33:39,395 INFO streaming.PipeMapRed: R/W/S=100/0/0 in:NA [rec/s] out:NA [rec/s]
2026-07-25 07:33:39,423 INFO streaming.PipeMapRed: Records R/W=900/1
2026-07-25 07:33:39,443 INFO streaming.PipeMapRed: MRErrorThread done
2026-07-25 07:33:39,445 INFO streaming.PipeMapRed: mapRedFinished
2026-07-25 07:33:39,451 INFO mapred.LocalJobRunner:
2026-07-25 07:33:39,451 INFO mapred.MapTask: Starting flush of map output
2026-07-25 07:33:39,452 INFO mapred.MapTask: Spilling map output
2026-07-25 07:33:39,452 INFO mapred.MapTask: bufstart = 0; bufend = 14400; bufvoid = 104857600
2026-07-25 07:33:39,452 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26210800(104843200); length = 3597/6553600
2026-07-25 07:33:39,530 INFO mapred.MapTask: Finished spill 0
2026-07-25 07:33:39,568 INFO mapred.Task: Task:attempt_local777582854_0001_m_000000_0 is done. And is in the process of committing
2026-07-25 07:33:39,579 INFO mapred.LocalJobRunner: Records R/W=900/1
2026-07-25 07:33:39,580 INFO mapred.Task: Task 'attempt_local777582854_0001_m_000000_0' done.
```



```
2026-07-25 07:33:39,590 INFO mapred.Task: Final Counters for attempt_local777582854_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=1677
                FILE: Number of bytes written=663736
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=21800
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=900
                Map output records=900
                Map output bytes=14400
                Map output materialized bytes=16206
                Input split bytes=118
                Combine input records=0
                Spilled Records=900
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=5
                Total committed heap usage (bytes)=203423744
        File Input Format Counters
                Bytes Read=21800
2026-07-25 07:33:39,591 INFO mapred.LocalJobRunner: Finishing task: attempt_local777582854_0001_m_000000_0
2026-07-25 07:33:39,591 INFO mapred.LocalJobRunner: map task executor complete.
2026-07-25 07:33:39,597 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-07-25 07:33:39,597 INFO mapreduce.Job: Job job_local777582854_0001 running in uber mode : false
2026-07-25 07:33:39,597 INFO mapred.LocalJobRunner: Starting task: attempt_local777582854_0001_r_000000_0
2026-07-25 07:33:39,599 INFO mapreduce.Job:  map 100% reduce 0%
2026-07-25 07:33:39,615 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:33:39,615 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:33:39,616 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-07-25 07:33:39,625 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@2c56c981
2026-07-25 07:33:39,628 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-07-25 07:33:39,666 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=688494976, maxSingleShuffleLimit=172123744, mergeThreshold=454406688, ioSor
tFactor=10, memToMemMergeOutputsThreshold=10
```

```
                FILE: Number of bytes read=34121
                FILE: Number of bytes written=679942
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=21800
                HDFS: Number of bytes written=38
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=9
                Reduce shuffle bytes=16206
                Reduce input records=900
                Reduce output records=1
                Spilled Records=900
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=6
                Total committed heap usage (bytes)=203423744
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=38
2026-07-25 07:33:40,146 INFO mapred.LocalJobRunner: Finishing task: attempt_local777582854_0001_r_000000_0
2026-07-25 07:33:40,147 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-07-25 07:33:40,602 INFO mapreduce.Job:  map 100% reduce 100%
2026-07-25 07:33:40,603 INFO mapreduce.Job: Job job_local777582854_0001 completed successfully
2026-07-25 07:33:40,625 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=35798
                FILE: Number of bytes written=1343678
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=43600
                HDFS: Number of bytes written=38
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=900
                Map output records=900
                Map output bytes=14400
                Map output materialized bytes=16206
                Input split bytes=118
```

```
2026-07-25 07:33:40,602 INFO mapreduce.Job:  map 100% reduce 100%
2026-07-25 07:33:40,603 INFO mapreduce.Job: Job job_local777582854_0001 completed successfully
2026-07-25 07:33:40,625 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=35798
                FILE: Number of bytes written=1343678
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=43600
                HDFS: Number of bytes written=38
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=900
                Map output records=900
                Map output bytes=14400
                Map output materialized bytes=16206
                Input split bytes=118
                Combine input records=0
                Combine output records=0
                Reduce input groups=9
                Reduce shuffle bytes=16206
                Reduce input records=900
                Reduce output records=1
                Spilled Records=1800
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=11
                Total committed heap usage (bytes)=406847488
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=21800
        File Output Format Counters
                Bytes Written=38
2026-07-25 07:33:40,625 INFO streaming.StreamJob: Output directory: /user/pratham/weather/output
```

```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hdfs dfs -cat /user/$USER/weather/output/part-00000
2026-07-25 07:33:46,094 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Hottest Day: 2024-01-09 with 33.0°C
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hdfs dfs -rm -r -skipTrash /user/$USER/weather/output_min
2026-07-25 07:33:52,376 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /user/pratham/weather/output_min
```

```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/weather/input/weather_large.txt \
-output /user/$USER/weather/output_min \
-mapper "python3 mapper_max_temp.py" \
-reducer "python3 reducer_min_temp.py" \
-file mapper_max_temp.py \
-file reducer_min_temp.py
2026-07-25 07:33:58,357 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
2026-07-25 07:33:58,497 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [mapper_max_temp.py, reducer_min_temp.py] [] /tmp/streamjob1806199931949624059 5.jar tmpDir=null
2026-07-25 07:33:59,045 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-07-25 07:33:59,175 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-07-25 07:33:59,175 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-07-25 07:33:59,191 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-07-25 07:33:59,656 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 07:33:59,755 INFO mapreduce.JobSubmitter: number of splits:1
2026-07-25 07:33:59,961 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1754778972_0001
2026-07-25 07:33:59,961 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 07:34:00,219 INFO mapred.LocalDistributedCacheManager: Localized file:/home/pratham/Hadoop_Exercise/mapper_max_temp.py as file:/tmp/hadoop-pratha
m/mapred/local/job_local1754778972_0001_58b2002e-f279-4756-9f42-27b3e32e9012/mapper_max_temp.py
2026-07-25 07:34:00,245 INFO mapred.LocalDistributedCacheManager: Localized file:/home/pratham/Hadoop_Exercise/reducer_min_temp.py as file:/tmp/hadoop-prath
am/mapred/local/job_local1754778972_0001_1ffd7b91-bafb-41c7-94e6-0abac44445a0/reducer_min_temp.py
2026-07-25 07:34:00,323 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-07-25 07:34:00,326 INFO mapreduce.Job: Running job: job_local1754778972_0001
2026-07-25 07:34:00,326 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-07-25 07:34:00,332 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-07-25 07:34:00,338 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:34:00,339 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:34:00,404 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-07-25 07:34:00,428 INFO mapred.LocalJobRunner: Starting task: attempt_local1754778972_0001_m_000000_0
2026-07-25 07:34:00,467 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:34:00,468 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:34:00,485 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-07-25 07:34:00,496 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/pratham/weather/input/weather_large.txt:0+21800
2026-07-25 07:34:00,517 INFO mapred.MapTask: numReduceTasks: 1
2026-07-25 07:34:00,650 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-07-25 07:34:00,650 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
```

```
2026-07-25 07:34:01,340 INFO mapreduce.Job: Job job_local1754778972_0001 completed successfully
2026-07-25 07:34:01,356 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=35786
                FILE: Number of bytes written=1349910
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=43600
                HDFS: Number of bytes written=38
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=900
                Map output records=900
                Map output bytes=14400
                Map output materialized bytes=16206
                Input split bytes=118
                Combine input records=0
                Combine output records=0
                Reduce input groups=9
                Reduce shuffle bytes=16206
                Reduce input records=900
                Reduce output records=1
                Spilled Records=1800
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=4
                Total committed heap usage (bytes)=425721856
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=21800
        File Output Format Counters
                Bytes Written=38
2026-07-25 07:34:01,356 INFO streaming.StreamJob: Output directory: /user/pratham/weather/output_min
```

```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hdfs dfs -cat /user/$USER/weather/output_min/part-00000
2026-07-25 07:34:07,616 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Coldest Day: 2024-01-03 with 10.0°C
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ cat > sales_products.txt << 'EOF'
2024-01-01,Laptop,45000,Delhi
2024-01-01,Mouse,1500,Mumbai
2024-01-02,Keyboard,3500,Bangalore
2024-01-02,Laptop,45000,Chennai
2024-01-03,Mouse,1500,Delhi
2024-01-03,Monitor,25000,Mumbai
2024-01-04,Laptop,45000,Bangalore
2024-01-04,Keyboard,3500,Delhi
2024-01-05,Monitor,25000,Chennai
2024-01-05,Mouse,1500,Mumbai
EOF
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ for i in {1..50}
do
cat sales_products.txt
done > sales_products_large.txt
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ mapper_top_products.py
mapper_top_products.py: command not found
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ chmod +x mapper_top_products.py
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ reducer_top_products.py
reducer_top_products.py: command not found
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ ls -l
total 72
drwxr-xr-x 2 pratham pratham  4096 Jul 25 07:22 Hadoop_Exercise
-rwxr-xr-x 1 pratham pratham   396 Jul 25 07:33 mapper_max_temp.py
-rwxr-xr-x 1 pratham pratham   387 Jul 25 07:19 mapper_top_products.py
-rwxr-xr-x 1 pratham pratham   377 Jul 25 07:33 reducer_max_temp.py
-rwxr-xr-x 1 pratham pratham   376 Jul 25 07:33 reducer_min_temp.py
-rwxr-xr-x 1 pratham pratham   770 Jul 25 07:19 reducer_top_products.py
-rw-r--r-- 1 pratham pratham   313 Jul 25 07:34 sales_products.txt
-rw-r--r-- 1 pratham pratham 15650 Jul 25 07:34 sales_products_large.txt
-rw-r--r-- 1 pratham pratham   218 Jul 25 07:32 weather_data.txt
-rw-r--r-- 1 pratham pratham 21800 Jul 25 07:32 weather_large.txt
```


```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ ls -l
total 72
drwxr-xr-x 2 pratham pratham  4096 Jul 25 07:22 Hadoop_Exercise
-rwxr-xr-x 1 pratham pratham   396 Jul 25 07:33 mapper_max_temp.py
-rwxr-xr-x 1 pratham pratham   387 Jul 25 07:19 mapper_top_products.py
-rwxr-xr-x 1 pratham pratham   377 Jul 25 07:33 reducer_max_temp.py
-rwxr-xr-x 1 pratham pratham   376 Jul 25 07:33 reducer_min_temp.py
-rwxr-xr-x 1 pratham pratham   770 Jul 25 07:19 reducer_top_products.py
-rw-r--r-- 1 pratham pratham   313 Jul 25 07:34 sales_products.txt
-rw-r--r-- 1 pratham pratham 15650 Jul 25 07:34 sales_products_large.txt
-rw-r--r-- 1 pratham pratham   218 Jul 25 07:32 weather_data.txt
-rw-r--r-- 1 pratham pratham 21800 Jul 25 07:32 weather_large.txt
```


```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/$USER/sales_products/input/sales_products_large.txt \
-output /user/$USER/sales_products/output \
-mapper "python3 mapper_top_products.py" \
-reducer "python3 reducer_top_products.py" \
-file mapper_top_products.py \
-file reducer_top_products.py
2026-07-25 07:38:14,875 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
2026-07-25 07:38:15,015 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [mapper_top_products.py, reducer_top_products.py] [] /tmp/streamjob16157297498151377039.jar tmpDir=null
2026-07-25 07:38:15,606 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-07-25 07:38:15,706 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-07-25 07:38:15,707 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-07-25 07:38:15,724 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-07-25 07:38:16,063 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 07:38:16,132 INFO mapreduce.JobSubmitter: number of splits:1
2026-07-25 07:38:16,285 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local274521064_0001
2026-07-25 07:38:16,285 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 07:38:16,506 INFO mapred.LocalDistributedCacheManager: Localized file:/home/pratham/Hadoop_Exercise/mapper_top_products.py as file:/tmp/hadoop-pr
atham/mapred/local/job_local274521064_0001_19abc783-c692-4198-a8ec-0b5b41459b1b/mapper_top_products.py
2026-07-25 07:38:16,514 INFO mapred.LocalDistributedCacheManager: Localized file:/home/pratham/Hadoop_Exercise/reducer_top_products.py as file:/tmp/hadoop-p
ratham/mapred/local/job_local274521064_0001_0f3d7293-d5e1-42ee-8c41-6a55806a2359/reducer_top_products.py
2026-07-25 07:38:16,574 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-07-25 07:38:16,576 INFO mapreduce.Job: Running job: job_local274521064_0001
2026-07-25 07:38:16,576 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-07-25 07:38:16,585 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-07-25 07:38:16,591 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:38:16,591 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:38:16,651 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-07-25 07:38:16,654 INFO mapred.LocalJobRunner: Starting task: attempt_local274521064_0001_m_000000_0
2026-07-25 07:38:16,673 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-07-25 07:38:16,674 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
2026-07-25 07:38:16,684 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-07-25 07:38:16,707 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/user/pratham/sales_products/input/sales_products_large.txt:0+15650
2026-07-25 07:38:16,719 INFO mapred.MapTask: numReduceTasks: 1
2026-07-25 07:38:16,803 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-07-25 07:38:16,803 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-07-25 07:38:16,805 INFO mapred.MapTask: soft limit at 83886080
```

```
                FILE: Number of write operations=0
                HDFS: Number of bytes read=31300
                HDFS: Number of bytes written=177
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=500
                Map output records=500
                Map output bytes=7400
                Map output materialized bytes=8406
                Input split bytes=132
                Combine input records=0
                Combine output records=0
                Reduce input groups=4
                Reduce shuffle bytes=8406
                Reduce input records=500
                Reduce output records=6
                Spilled Records=1000
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=18
                Total committed heap usage (bytes)=419430400
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=15650
        File Output Format Counters
                Bytes Written=177
2026-07-25 07:38:17,598 INFO streaming.StreamJob: Output directory: /user/pratham/sales_products/output
```

```
pratham@LAPTOP-PL4G895R:~/Hadoop_Exercise$ hdfs dfs -cat /user/$USER/sales_products/output/part-00000
2026-07-25 07:38:23,040 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
TOP 5 PRODUCTS BY SALES
========================================
1. Laptop: ₹6,750,000.00
2. Monitor: ₹2,500,000.00
3. Keyboard: ₹350,000.00
4. Mouse: ₹225,000.00
```

Conclusion

This project demonstrates the practical implementation of the Hadoop MapReduce framework for processing large-scale datasets. By using Python-based Mapper and Reducer programs, weather data was analysed to determine the hottest and coldest days, while sales data was processed to identify the top-selling products and city-wise product performance. The project highlights the efficiency of distributed computing in handling big data and shows how Hadoop Streaming can be integrated with Python to perform scalable data analysis. Overall, the exercises provide a strong understanding of MapReduce programming, HDFS operations, and real-world big data analytics using Hadoop.